Credit Card Fraud Detection — Business Analysis Case Study

Overview

- **Role:** Business Analyst (Academic Project)
- **Context:** Academic project focused on applying business analysis practices to a fraud detection problem
- **Tools Used:** Jira, Notion, Lucidchart, Python (Scikit-Learn)
- **Deliverables:** Requirements documentation, process flow diagram, model evaluation, and business recommendations

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1) Business Problem

Financial institutions process a high volume of transactions, making it difficult to accurately detect fraudulent activity, especially due to class imbalance. The goal is to improve fraud detection while minimizing false positives and operational impact.

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2) Stakeholders

- **Fraud Analysts**
  Responsible for reviewing flagged transactions and confirming fraudulent activity
- **Operations Team**
  Handles transaction processing and manages flagged or blocked transactions
- **Customers**
  Impacted by fraudulent activity and false positives affecting legitimate transactions
- **IT / Development Team**
  Responsible for implementing and maintaining the fraud detection system

- **Business Stakeholders (Management)**
  Focused on reducing financial loss and improving operational efficiency
- **Project Sponsor (Instructor)**
  Provides project direction, requirements, and evaluation

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3) Business Requirements (Functional and Non-Functional)

Functional requirements

- The system must evaluate transaction data to identify potential fraud
- The system must assign a fraud risk score to each transaction
- The system must flag high-risk transactions for further review
- The system must allow analysts to review flagged transactions

Non-functional requirements

- The system should achieve high accuracy in detecting fraudulent transactions
- The system should minimize false positives to avoid disrupting legitimate transactions
- The system should process transactions efficiently for near real-time detection
- The system should scale to handle large volumes of transaction data

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4) Epics & User Stories

Epics

- **Requirements & Stakeholder Analysis**
- **Process Flow & Workflow Design**
- **Model Evaluation & Analysis**
- **Recommendations & Documentation**

Sample user stories

- As a fraud analyst, I want to identify high-risk transactions so that fraudulent activity can be prevented
- As an operations team member, I want transactions to be accurately flagged so that legitimate transactions are not disrupted
- As a business stakeholder, I want an effective fraud detection approach so that financial loss is minimized

Acceptance criteria

- Fraud detection requirements clearly defined
- Risk scoring logic documented
- Process flow accurately represents system behavior
- Model evaluation supports business decision-making

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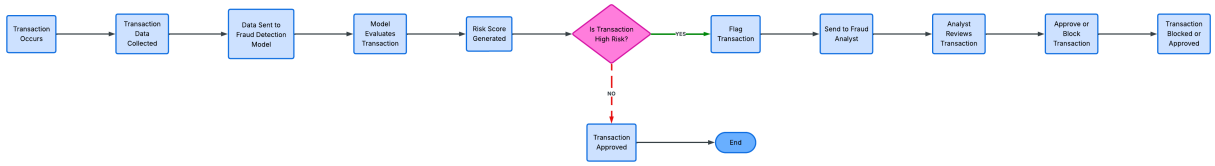
5) Process Flow / Workflow



> Figure 1: Fraud Detection Process Flow

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6) Data Analysis & Key Findings

The dataset used for fraud detection is highly imbalanced, with only 0.17% of transactions labeled as fraudulent. This imbalance makes traditional accuracy metrics misleading, as models may achieve high accuracy while failing to detect fraudulent cases effectively.

To address this, performance was evaluated using precision, recall, and F1 score, which provide a better understanding of how well the models detect fraud. Recall

is especially important in this context, as it measures the ability to correctly identify fraudulent transactions.

The analysis revealed that different models performed well under different conditions, highlighting the importance of balancing detection performance with operational efficiency.

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7) Solution Evaluation (Model Comparison)

Several models were evaluated, including Random Forest, AdaBoost, Perceptron, and K-Nearest Neighbors (KNN).

- **Random Forest** provided strong overall performance with a good balance between accuracy and runtime
- **KNN** achieved high performance metrics but had slower prediction times, making it less suitable for real-time applications
- **AdaBoost** offered faster runtime but lower recall, making it less effective for detecting fraud
- **Perceptron** was the fastest but had the weakest overall performance

Each model demonstrated trade-offs between performance, speed, and practicality, which were considered in determining the best solution.

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8) Trade-Off Analysis

Several trade-offs were identified when evaluating the fraud detection models. While some models provided higher accuracy and performance metrics, they were often slower and less practical for real-time applications.

KNN demonstrated strong performance in terms of precision and F1 score, but its slower prediction time makes it less suitable for high-speed transaction environments. On the other hand, Random Forest offered a strong balance between performance and runtime, making it more practical for real-time fraud detection.

Additionally, the choice of evaluation metrics played a critical role. Accuracy alone was not a reliable measure due to the class imbalance in the dataset. Metrics such

as recall and F1 score were prioritized to ensure fraudulent transactions were effectively detected.

These trade-offs highlight the importance of balancing detection performance with operational efficiency.

9) Recommendations

Based on the analysis, Random Forest is recommended as the primary model for fraud detection due to its balance between performance, speed, and reliability.

While KNN achieved strong performance metrics, its slower runtime limits its effectiveness in real-time scenarios. Random Forest provides a more scalable and efficient solution while maintaining strong detection capability.

To further improve the system, future enhancements could include using a more balanced dataset, implementing additional feature engineering, and exploring ensemble or hybrid approaches to improve detection accuracy.

These recommendations align with business goals of reducing fraud, minimizing false positives, and maintaining operational efficiency.

10) Key Outcomes

- Defined business requirements for fraud detection in a high-risk, imbalanced dataset environment
- Translated model performance metrics into business-focused insights and recommendations
- Designed a process flow representing system behavior and decision points
- Applied Agile practices using Jira to manage work and track progress

11) Agile Approach

The project was structured using an Agile approach in Jira, organizing work into sprints and backlog items.

- **Sprint 1:** Business problem definition, stakeholder identification, and requirements documentation

- **Sprint 2:** Process flow and workflow mapping
- **Sprint 3:** Model evaluation and analysis
- **Sprint 4:** Trade-off analysis and final recommendations

Work was tracked using Jira with stories moving through **To Do, In Progress, Review, and Done** states.

This iterative approach allowed for continuous refinement of requirements, analysis, and recommendations based on insights gained throughout the project.

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12) Conclusion

This project demonstrates the application of business analysis principles to a fraud detection problem, including requirements gathering, process mapping, data analysis, and solution recommendation.

By translating technical model performance into business-focused insights, the project highlights the importance of balancing detection accuracy, operational efficiency, and real-world usability.

The use of Agile practices and Jira further demonstrates the ability to structure work, prioritize tasks, and deliver outcomes in an iterative, organized manner.

Appendix

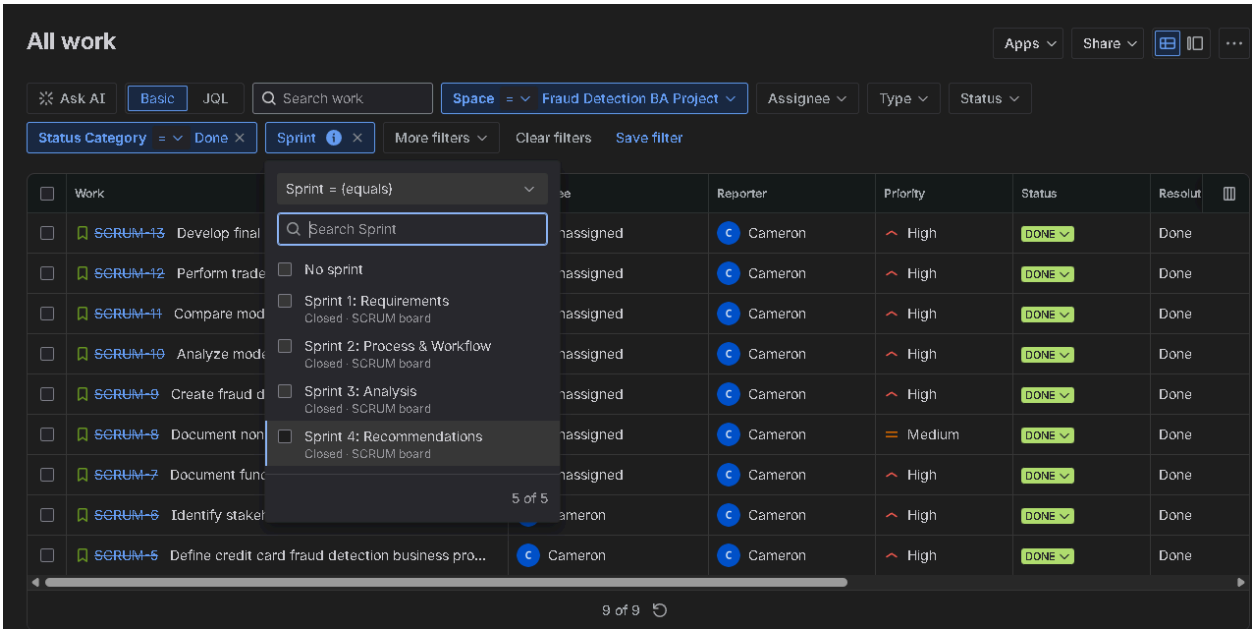

Figure 2: Jira sprint structure illustrating work organized into four Agile sprints covering requirements, process design, analysis, and final recommendations.

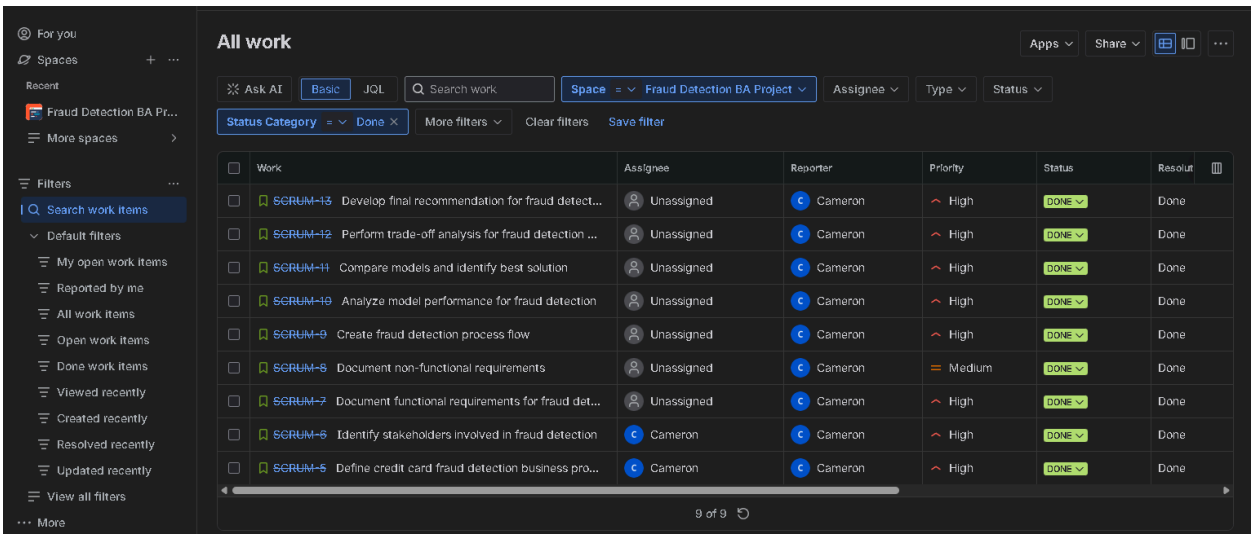

Figure 3: Completed work items tracked in Jira after sprint completion